

Brighton D

bright851999@gmail.com

+91 99868 25798

Research Associate · Computer Vision & AI Engineer

[linkedin.com/in/brighton-d](https://www.linkedin.com/in/brighton-d)

PROFILE

Research Associate at Rangsons Aerospace Pvt. Ltd. (RAPL) with 1+ years of industry experience and research interests in Computer Vision, Deep Learning, Image Processing, and Geo-Intelligence. Experienced in developing AI-based image analysis systems for aerial and remote sensing imagery using PyTorch, TensorFlow, OpenCV, and CUDA-enabled frameworks. Conducted research and development on hybrid oriented object detection approaches for aerial and remote sensing imagery. Strong foundation in mathematical modelling, scientific computing, and machine learning, complemented by research publication experience in Computer Vision. Building expertise in enterprise AI and agentic AI systems.

EMPLOYMENT HISTORY

Research Associate Sep 2025 – Present
Rangsons Aerospace Pvt. Ltd. (RAPL), Bengaluru

- Architected and led a 5-person team building an agentic RAG platform for multi-format enterprise data analysis and insight generation, deployed as containerized Docker microservices
- Leading a 5-person team building TrainForge AI, a platform for dataset annotation, multi-model training, registration, and inferencing
- Delivered a hybrid oriented object detection approach individually for an aerospace & defence client, achieving 78%+ mAP and producing two research manuscripts as client technical deliverables
- Worked at the Centre for Airborne Systems (CABS), Sep 2025 — Apr 2026, on AI-based computer vision for aerial image analysis
- Designed preprocessing and annotation pipelines across multi-disciplinary teams

Research & Data Analytics Intern May – Jun 2019
Vajra Global Consulting Services LLP, Chennai

- Organized and analyzed revenue-related data from client accounts
- Translated data insights into structured reports for internal presentation

EDUCATION

Ph.D., Mathematics Apr 2022 – Aug 2025
Madras Christian College, Chennai

M.Sc., Mathematics Jul 2019 – Apr 2021
Madras Christian College, Chennai

B.Sc., Mathematics
Madras Christian College, Chennai

Jun 2016 – Apr 2019

HSC (ICE)
King of Kings Matric. Hr. Sec. School, Tuticorin

2016

SSLC
BMC Matriculation Hr. Sec. School, Tuticorin

2014

PUBLICATIONS

Brighton, D., Thamburaj, R., Sheeba, F., Richard, P.S.S. (2025). Classification of Pollen Grains Using Convolutional Neural Networks. CIS 2024, LNNS vol 1275, Springer, Singapore.

Brighton, D., Thamburaj, R., Sheeba, F., Richard, P.S.S. (2025). Segmentation Techniques in Identifying Cavities Due to Stem Borer Beetles: A Study on Images of *Mangifera indica* L. cv. mallika. CompCom 2025, LNNS vol 1425, Springer, Cham.

CERTIFICATIONS

Python Programming for Engineering Applications — OOP, error handling, regex, multithreading/multiprocessing, chatbot development, LLMs and RAG pipelines

Excel Power Tools for Data Analytics — Power Query, Power Pivot, automated reporting

Google Analytics — digital marketing analytics, user behavior, data-driven decisions

SKILLS

Programming: Python

Deep Learning: PyTorch, TensorFlow, Keras, MMRotate

Computer Vision: Object Detection, Oriented Object Detection, CNNs, Transformer Architectures, Image Segmentation

Image Processing: OpenCV, Image Augmentation, Feature Extraction, Annotation Handling

Remote Sensing: Aerial Image Analysis, Geospatial Image Analysis, Geo-Intelligence

Scientific Computing: NumPy, Pandas, Mathematical Modeling

GenAI / Agentic: Agentic RAG, LLM APIs, Prompt Engineering, Vector Databases

Tools: Docker, CUDA, Git, Linux

LEADERSHIP & ACTIVITIES

Student Placement Coordinator — M.Sc. Mathematics (2020–2021): liaison between students and recruiting firms, coordinated placement drives and soft-skill training

Treasurer — Mathematics Association (2019–2020): managed annual budget, allocated resources for academic events

NSS Leader — National Service Scheme (2018–2019): led a 7-day rural outreach program on health, sanitation, and community education

LANGUAGES

Tamil — Native Speaker · English — Proficient